Harbourline Foods demand planner: a seven-week path from forecast to order

A working prototype of a grocery demand-forecasting pilot: models backtested in dollars, a planner's order screen, an earned autonomy ladder and the business case that decides whether to go on.

The question

Harbourline Foods is a fictional mid-sized Canadian grocer: 300 stores, about $200M in revenue, and weekly ordering that leans on last year's sales. Too little costs a sale; too much costs waste and cash. Can a better forecast tell each store how much to order, and does the gain clear a go/no-go line of $0.70M a year in profit?

The approach

Discovery. Seven weeks with three checkpoints where the work stops if the numbers do not clear: week 2, can lost demand be told from empty shelves; week 5, does a model beat the baseline; week 7, do the numbers justify going ahead. Six risks each carry a named owner.

Models. Five candidates on 104 weeks of sales for 20 pilot stores and 136 products in 8 categories, from same week last year to gradient-boosted trees with a Poisson loss and a hierarchical reconciliation. Each was tested on a rolling 12-week holdout and judged in dollars: lost margin, waste and carrying cost. The selection rule picks the simplest model within 10% of the best.

Planner. One screen per store shows next week and the week after, a P10 to P90 range, stock position and a suggested order in whole cases. Seven review rules decide what needs a planner; everything else is approved automatically, so a store reviews 15% to 20% of its lines. Every override carries a reason code.

Autonomy. Trust is earned per category and can be lost. L1 approves every line, L2 reviews exceptions only, L3 writes the orders with a planner veto. Eight passing weeks earn promotion, two consecutive misses demote, and fresh stays at L2 through the pilot.

Business case. Three levers set at or below the low end of published vendor results: sell more, throw out less, hold less stock. Base case $1.06M a year against a $0.70M break-even, payback 2.7 years, with $0.175M of discovery buying three decisions before the $0.45M pilot and $0.60M rollout.

The outcome

Metric	Baseline	Selected model
Error next week (WAPE)	39.7%	28.7%
12-week cost of forecast error	$408K	$319K
Saved against baseline		$89K
In-stock across 20 stores		93%
Spoilage at cost, share of sales		1.5%

Hierarchical reconciliation scored 28.6% and $91K, not enough to justify the extra moving part. Synthetic data is cleaner than real data, so the results are illustrative.

What we would do next

- Estimate lost demand from stock records, so the baseline is demand, not sales.
- Forecast daily for produce, bakery and meat.
- Connect to the ordering system for real through the EDI loop retailers already run.
- Pilot in 20 to 30 stores, in shadow mode first.

What this demonstrates

- **Discovery craft.** A value tree, three checkpoints and six owned risks, so a sponsor can say stop as easily as go.

- **Honest evaluation.** Models judged in dollars; the simplest adequate model chosen; the synthetic uplift labelled as such.
- **Adoption designed in.** Review by exception, a reason on every override, and an autonomy ladder that earns and loses trust on evidence.
- **Stage-gated spend.** Discovery, pilot and rollout funded in turn, each unlocked by a number agreed in advance.
- **AI-native build with human ownership.** About two working days of elapsed time with Claude Code and Codex; 30 Python and 7 Node tests, a deterministic pipeline and CI on every push, with the author reviewing every model choice, threshold and sentence.

Live: https://harbourline-demand-planner.vercel.appSMARTPANTS_PRESERVED_126 · Code: https://github.com/ankitmishra01/harbourline-demand-plannerSMARTPANTS_PRESERVED_128 · Harbourline Foods is a fictional retailer; all data is synthetic and calibrated to public Statistics Canada series. Not affiliated with any company. Built by Ankit Mishra.